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Control Allocation of Rear Wheel Steering and Electronic Stability Control with Actuator Failure

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***Abstract*— This paper has presented optimal coordination of Rear Wheel Steering and Electronic Stability Control. The algorithm consists of three parts. First, state estimators for tire force, cornering stiffness and side slip angle have been designed. Second, desired vehicle motion has been generated considering driving situation and driver's intention. Third, optimal coordinator has distributed the control task to the individual chassis control using analytical method with Karush-Kuhn-Tucker. The purpose of this study can be divided into two categories. One is optimal control of Rear Wheel Steering and Electronic Stability Control. Therefore, new cost functions have been defined. The other is fault tolerant control during actuator fault or brake circuit failure. The performance of the proposed control system has been validated through closed-loop simulations with MATLAB/Simulink and Carsim. The simulation results show that proposed system does not only distribute the control input to each control system but also ensure vehicle safety under actuator failure.**

***Keywords*—integrated chassis control, fault tolerant control, rear wheel steering, electronic stability control, control allocation**

I. INTRODUCTION

To improve vehicle driving performance and stability, various chassis control systems have been developed over the last 20 years. Among those systems, Electronic Stability Control (ESC) is well known as an effective system for ensuring vehicle stability using differential braking. In addition, there are various individual chassis control systems such as Torque Vectoring (TV), Active Front Steering (AFS), Rear Wheel Steering (RWS), Anti Roll System (ARS) and Electronic Control Suspension (ECS). Each system is developed to improve handling performance, lateral stability and driving comfort [13]. For maximizing stability and driving performance, the integration of individual modular chassis control systems has been studied by several researchers. Previous researches have shown that both stability and driving performances can be significantly improved with integrated chassis control system [6]-[8], [14], [15].

For ensuring handling steerability, integration of Variable Gear Ratio Steering (VGS) and RWS has been proposed [1]. Conventional RWS actuator controls the rear steering angle for improving vehicle lateral stability through reducing the vehicle side slip angle. Due to these characteristic of RWS, side slip angle and steerability can be improved at low speed, however, it is difficult to improve steerability at high speed. To overcome these limitation, integrated steering control of VGS and RWS has been researched. However, it is difficult to ensure expected performance in critical driving situation. Accordingly, coordination of brake control and steering control system has been proposed [3]. For improving lateral stability, maneuverability and agility, the unified chassis control of AFS and ESC has been developed. According to driving situation and driver intention, desired vehicle motion is determined. Proposed control system can minimize the unexpected ESC braking intervention and improve performance in various driving situation. Proposed chassis control system has shown that handling performance and lateral stability can be improved more effective than individual chassis control system. Also, coordination of Four Wheel Steering (4WS) and Four Wheel Driving (4WD) has been reported [2], [4]. To minimize the tire usage, each longitudinal tire force and lateral tire force is distributed optimally. These researches have shown that handling, driving and braking performance can be dramatically increased using optimization method. In contrast, the risk of failure increases when various chassis control systems are equipped on one vehicle. Especially, previous research has reported that Steer By Wire (SBW) system such as AFS and RWS can lead to critical accident when SBW failure occurs [5]. For example, undesired yaw moment is generated suddenly when steering wheel actuator fails. In this situation, it is difficult to overcome undesired yaw moment using counter steer by driver. Also, if brake system has failure, the vehicle can be lead to serious dangerous situation. Recently, some researchers have studied fault tolerant control system based on redundancy characteristic of integrated chassis control system [16]-[18].

This paper presents integrated chassis control of RWS and ESC. The target vehicle of this study is equipped with hydraulic brake system with ESC and RWS that controls rear steering angle with the same amount using an actuator motor. The objectives of this paper can be divided into two parts as follows.

1. Coordinated control of RWS and ESC considering driver's braking intention and driving situation is performed with analytical optimization method.
2. Fault tolerant control algorithm is proposed. For example, when RWS actuator fault or brake circuit failure occurs, proposed control system can ensure the stability and safety.

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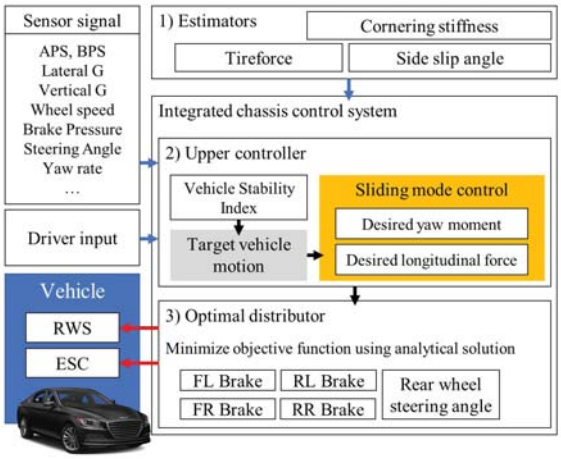

Fig. 1. Overall Architecture of proposed integrated chassis control

The performance of the proposed control system is investigated via closed-loop simulations with commercial software MATLAB/Simulink and Carsim [20].

II. Integrated Chassis Control Arhchitecture

Integrated chassis control system consists of the three parts: State estimator, Upper controller and Optimal distributor. Proposed integrated chassis control is described in Fig. 1.

1) State estimator: Information that cannot be directly measured from the sensors is estimated.

2) Upper controller: Through the sensor signals and estimated information, target vehicle motion is generated. After that, desired yaw moment and longitudinal force are determined using Sliding Mode Control (SMC) for tracking desired vehicle motion.

3) Optimal distributor: Each brake moment and the rear steering angle is coordinated using analytical optimization solution. Also, inequality condition is considered using Karush-Kuhn-Tucker (KKT) conditions.

III. State Estimation

For optimal coordination of individual chassis control, vehicle state and driving status should be estimated reasonably. However, it is difficult to estimate that information precisely. In this paper, cornering stiffness, side slip angle and 3-axis tire forces are estimated.

Tire force estimator is implemented by referring to previous research [9]. In short, vertical, lateral and longitudinal tire force are estimated following several steps. First, each vertical tire force is calculated. Second, wheel shaft torque is estimated by kalman filter. Third, each longitudinal tire force is estimated using Sliding Mode Observer (SMO). Finally, each lateral tire force is estimated based on extended kalman filter. Estimated tire force values are key information to solve the optimization problem.

Side slip angle β is an indicator of vehicle stability. For side slip angle estimation, model parameters such as cornering

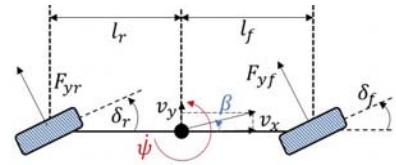

Fig. 2. 2-DOF bicycle model

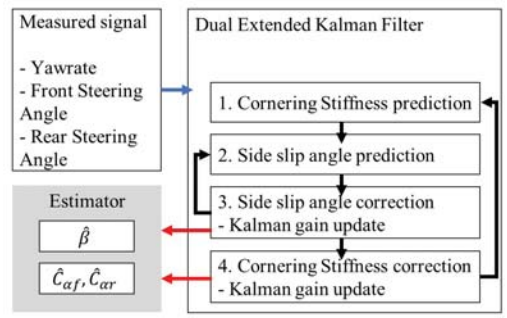

Fig. 3. Scheme of DEKF

stiffness must be estimated simultaneously [10]. In this paper, cornering stiffness $C_{\alpha f}, C_{\alpha r}$ and side slip angle are estimated with same sampling time using Dual Extended Kalman Filter (DEKF) based on 2-DOF model in Fig. 2. Side slip angle and cornering stiffness estimation scheme is described in Fig. 3.

A discrete dynamic model can be formulated as following (1)-(4). Q, R and P are noise covariance, output noise covariance and covariance matrices of the estimation errors, respectively. Also, m and I_z are mass and yaw moment inertia.

1) Cornering stiffness prediction

$$
\begin{aligned}
&(x_p)_{k+1} = (I + A_p T)\cdot(x_P)_k \\
&(P_p)_{k+1} = (I + A_p T)\cdot(P_p)\cdot(I + A_p T)^T + Q_p \\
&\textit{where,} \\
&I = diag(1,1),\ T = sample\ time \\
&x_p = [C_{\alpha f}, C_{\alpha r}]',\ A_p = zeros(2,2)
\end{aligned} \tag{1}
$$

2) State prediction

$$
\begin{aligned}
&(x_s)_{k+1} = (I + A_s T)\cdot(x_s)_k + (B_s T)\cdot u \\
&(P_s)_{k+1} = (I + A_s T)\cdot(P_s)_k \cdot (I + A_s T)^T + Q_s \\
&\textit{where,} \\
&x_s = [\beta, \gamma]', u = [\delta_f, \delta_r] \\
&A_s = [(-2C_{\alpha f} - 2C_{\alpha r}) / mv_x, 2C_{\alpha r} l_r - 2C_{\alpha f} l_f) / (mv_x^2) - 1; \\
&(2C_{\alpha r} l_r - 2C_{\alpha f} l_f) / I_z, (-2C_{\alpha f} l_f^2 - 2C_{\alpha r} l_r^2) / (I_z v_x)] \\
&B_s = [2C_{\alpha f} / mv_x, 2C_{\alpha r} / mv_x; 2C_{\alpha f} l_f / I_z, -2C_{\alpha r} l_r / I_z]
\end{aligned} \tag{2}
$$

3) *State correction*

$$
\begin{aligned}
K_s &= (P_s)_{k+1} \cdot H^T (H \cdot (P_s)_{k+1} \cdot H^T + R_s)^{-1} \\
\hat{x}_s &= (x_s)_{k+1} + K_s (z - H \cdot (x_s)_{k+1}) \\
\hat{P}_s &= (P_s)_{k+1} - K_s \cdot H \cdot (P_s)_{k+1}
\end{aligned}
\tag{3}
$$

where,

$H = [0,1]$

$z = measured\ yawrate$

4) *Cornering stiffness correction*

$$
\begin{aligned}
K_p &= (P_p)_{k+1} \cdot C_P^{\ T} (C_P \cdot (P_p)_{k+1} \cdot C_P^{\ T} + R_P)^{-1} \\
\hat{x}_p &= (x_p)_{k+1} + K_p (z - H \cdot (x_s)_{k+1}) \\
\hat{P}_p &= (P_p)_{k+1} - K_p \cdot C_P \cdot (P_p)_{k+1}
\end{aligned}
\tag{4}
$$

where,

$\mathrm{e} = z - H \cdot (x_s)$

$C_P = -\dfrac{\partial e}{\partial x_p} = H \dfrac{\partial (x_s)_{k+1}}{\partial x_P} |_{x_p = (x_p)_{k+1}}$

Although DEKF requires high computation effort, it is robust to the noise of the measured sensor signal and it can estimate side slip angle and cornering stiffness according to the driving situation through tuning of R, Q and P matrices.

IV. UPPER CONTROLLER

A. Target vehicle motion

Desired vehicle motion can be determined according to the driving situation. In this research, vehicle stability index is generated to determine desired vehicle motion. Vehicle stability index is defined using the phase plane for $\beta - \dot{\beta}$ as following (5), [8].

$$I_{stabillity} = | a\beta + b\dot{\beta} | \tag{5}$$

Fig. 4 shows the meaning of vehicle stability index $I_{stabillity}$. It is a dimensionless number that can indicate the danger of stability due to a large side slip angle. It means that $I_{stabillity}$ is used for switching between the lateral stability control and handling performance control modes. For example, the index exceeds the threshold, vehicle should be controlled for lateral stability. To avoid chattering effects caused by the frequent value changes, hysteresis structure is designed for mode change function.

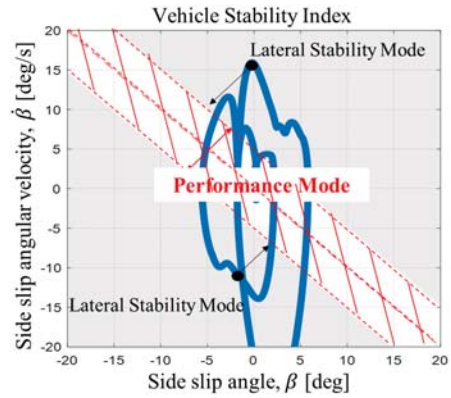

Fig. 4. Vehicle stability index

In this paper, desired yaw rate γ_{des} for cornering performance is generated using steady state turning condition (6). R_t, k_v and a_y are turning radius, understeer gradient and lateral acceleration, respectively [13].

$$
\begin{aligned}
\delta_{ss} &= \frac{L}{R_t} + k_v a_y \\
\gamma_{des,ss} &= \frac{\dot{x}}{R_t} = \frac{\dot{x}}{l_f + l_r + \dfrac{m\dot{x}^2 (l_r C_{\alpha r} - l_f C_{\alpha f})}{2 C_{\alpha f} C_{\alpha r} (l_f + l_r)}} \delta_f
\end{aligned}
\tag{6}
$$

$(\gamma_{des,ss}\ should\ be\ less\ than\ \gamma_{\max} = \dfrac{\mu g}{v_x})$

On the other hand, desired yaw rate for lateral stability is generated using Lyapunov law from 3-DOF model (7), Fig. 5.

$$
\begin{aligned}
m\dot{v}_y &= (F_{y,LF} + F_{y,RF}) \cos\delta_f + (F_{y,LR} + F_{y,RR}) \cos\delta_r - m v_x \gamma \\
\dot{\beta} &= \frac{(F_{y,LF} + F_{y,RF}) \cos\delta_f + (F_{y,LR} + F_{y,RR}) \cos\delta_r}{m v_x} - \gamma
\end{aligned}
$$

$$Lyapunov\ law\text{:}\ \dot{\beta} = -k\beta \tag{7}$$

$$
\begin{aligned}
&Thus, \gamma_{des,stability} \\
&= \{(F_{y,LF} + F_{y,RF}) \cos\delta_f + (F_{y,LR} + F_{y,RR}) \cos\delta_r\} / m v_x + k\beta
\end{aligned}
$$

In order to evaluate driver's acceleration and deceleration intention, desired longitudinal velocity v_x is generated using Accelerator Pedal Sensor (APS), Brake Pedal Sensor (BPS) (8).

$$v_{x,des} = v_x + \int f(APS, BPS) d\tau \tag{8}$$

B. Upper controller

The Sliding Mode Control scheme is adopted in this controller. Upper controller generates the desired longitudinal braking force and yaw moment to meet the driver's braking and steering intention considering uncertainty of the cornering stiffness. SMC based on 2-DOF and 3-DOF model can be designed as following [12]. λ, k_1, k_2 are positive constants.

1) SMC for determining the desired yaw moment, M^*

$$
\begin{aligned}
&S_1 = (\gamma - \gamma_{des}) + \lambda \int (\gamma - \gamma_{des}) d\tau, \ (\lambda > 0) \\
&\dot{s}_1 = (\dot{\gamma} - \dot{\gamma}_{des}) + \lambda(\gamma - \gamma_{des}) = 0 \\
&\left(\frac{2C_{\alpha r} l_r - 2C_{\alpha f} l_f}{I_z v_x} \right) v_y - \left(\frac{2C_{\alpha f} l_f^2 + 2C_{\alpha r} l_r^2}{I_z v_x} \right) \gamma \\
&+ \frac{2 l_f C_{\alpha f}}{I_z} \delta_f - \frac{2 l_r C_{\alpha r}}{I_z} \delta_r + \frac{M_{eq}^*}{I_z} - \dot{\gamma}_{des} + \lambda(\gamma - \gamma_{des}) \\
&+ disturbance = 0
\end{aligned} \tag{9}
$$

The equivalent input is,

$$
\begin{aligned}
&M_{eq}^* = -\left(\frac{2C_{\alpha r} l_r - 2C_{\alpha f} l_f}{v_x} \right) v_y + \left(\frac{2C_{\alpha f} l_f^2 + 2C_{\alpha r} l_r^2}{v_x} \right) \gamma \\
&-2 l_f C_{\alpha f} \delta_f + 2 l_r C_{\alpha r} \delta_r + I_z \dot{\gamma}_{des} - \lambda I_z (\gamma - \gamma_{des}) \\
&Thus, M^* = M_{eq}^* - k_1 \cdot sign(s_1)
\end{aligned}
$$

2) SMC for determining the desired braking force, F_x^*

$$
\begin{aligned}
&S_2 = (v_x - v_{x,des}) \\
&\dot{s}_2 = (\dot{v}_x - \dot{v}_{x,des}) = 0 \\
&\frac{1}{m}(F_{x,LF} + F_{x,RF}) cos\delta_f + \frac{1}{m}(F_{x,LR} + F_{x,RR}) cos\delta_r \\
&- \frac{1}{m}(F_{y,LF} + F_{y,RF}) sin\delta_f - \frac{1}{m}(F_{y,LR} + F_{y,RR}) sin\delta_r \\
&+ \gamma v_y + \frac{1}{m} F_x^* - \dot{v}_{x,des} + disturbance = 0
\end{aligned} \tag{10}
$$

The equivalent input is,

$$
\begin{aligned}
&F_{x,eq}^* = -(F_{x,LF} + F_{x,RF}) cos\delta_f - (F_{x,LR} + F_{x,RR}) cos\delta_r \\
&+ (F_{y,LF} + F_{y,RF}) sin\delta_f + (F_{y,LR} + F_{y,RR}) sin\delta_r - m(\gamma v_y - \dot{v}_{x,des}) \\
&Thus, F_x^* = F_{x,eq}^* - k_2 \cdot sign(s_2) \\
&(F_x^* \ should \ be \ less \ than \ zero)
\end{aligned}
$$

The Lyapunov function candidates $V_{1,2}$ are defined as $V_{1,2} = 0.5 \cdot s_{1,2}^2$. Design parameters $k_{1,2}$ can be designed bigger than disturbance. Therefore, the derivative of the Lyapunov functions $\dot{V}_{1,2} = s_{1,2} \dot{s}_{1,2}$ can be guaranteed to be stable (11).

$$
\begin{aligned}
&s_i \dot{s}_i = s(-k_i \cdot sign(s_i) + disturbance) = -\eta_i \mid s_i \mid \le 0 \\
&where, i = 1, 2
\end{aligned} \tag{11}
$$

In order to attenuate high-frequency chattering, a saturation functions, $sat(s_{1,2} / \Phi)$, are implemented. Finally, Each desired yaw moment and braking force can be formulated as $M^* = M_{eq}^* - k_1 \cdot sat\left(\frac{s_1}{\Phi_1}\right)$ and $F_x^* = F_{x,eq}^* - k_2 \cdot sat\left(\frac{s_2}{\Phi_2}\right)$, respectively. Desired yaw moment and braking force are inputs to lower controller, optimal distributor.

V. OPTIMAL DISTRIBUTOR

Based on the desired longitudinal force and yaw moment, optimal coordinator distributes control task to a steering actuator and each wheel brake. Fig. 5 shows the 3-DOF model and configuration of RWS and ESC. In general, the optimization problem can be divided into two methods. One is numerical method that shows great performance in general scenario. However, numerical method needs high computation effort and it would not converge in the given time. The other is an analytical method using Lagrange function [11]. In this paper, the optimization problem is solved by analytical method with Karush-Kuhn-Tucker (KKT) condition considering actuator constraints. Finally, analytical solution is applied to the distribution algorithm.

A. Variable Elimination

To solve the optimization problem with 6 variables ($F^*_{x,LF}, F^*_{x,LR}, F^*_{x,RF}, F^*_{x,RR}, F^*_{y,LR}, F^*_{y,RR}$), it is necessary to eliminate 3 variables ($F^*_{x,LR}, F^*_{x,RR}, F^*_{y,RR}$) based on certain assumptions. First, the active lateral force of right rear tire $F^*_{y,RR}$ can be determined as (12), [3].

$$
\begin{aligned}
&F^*_{y,RR} = Q_1 \cdot F^*_{y,LR} \\
&where, Q_1 = \frac{F_{z,RR}}{F_{z,\mathrm{LR}}} \\
&F_z \ is \ vertical \ tire \ force
\end{aligned} \tag{12}
$$

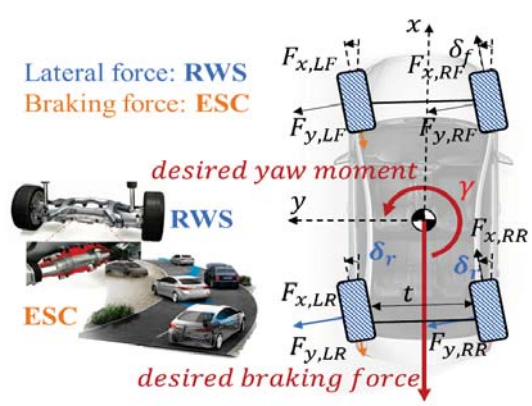

Fig. 5. 3-DOF planar model and Configuration of RWS and ESC

Second, the maximum brake forces of each tire can be evaluated using estimated tire forces (13), and the braking force distribution of the front and rear tires is determined. Each active rear braking force $F^*_{x,LR}, F^*_{x,RR}$ is presented as (14).

$$\begin{aligned}
F^*_{x,LF,max} &= -F_{x,LF} - \sqrt{(\mu \cdot F_{z,LF})^2 - (F_{y,LF})^2} \\
F^*_{x,RF,max} &= -F_{x,RF} - \sqrt{(\mu \cdot F_{z,RF})^2 - (F_{y,RF})^2} \\
F^*_{x,LR,max} &= -F_{x,LR} - \sqrt{(\mu \cdot F_{z,LR})^2 - (F_{y,LR})^2} \\
F^*_{x,RR,max} &= -F_{x,RR} - \sqrt{(\mu \cdot F_{z,RR})^2 - (F_{y,RR})^2} \\
&\textit{where}, \mu \textit{ is road friction coefficient}
\end{aligned} \tag{13}$$

$$\begin{aligned}
F^*_{x,LR} &= P_L \cdot F^*_{x,LF} \\
F^*_{x,RR} &= P_R \cdot F^*_{x,RF} \\
&\text{where,} \\
P_L &= \frac{|F^*_{x,LR,\max}|}{|F^*_{x,LF,\max}|}, P_R = \frac{|F^*_{x,RR,\max}|}{|F^*_{x,RF,\max}|}
\end{aligned} \tag{14}$$

B. Objective Function and Constraints

In a conventional ESC, the desired yaw moment is generated using differential braking. Consequently, unexpected deceleration is occurred. However, integration control of RWS and ESC can minimize undesired deceleration to meet desired yaw moment considering driver's intention. It is because RWS can generate yaw moment using rear steering. Furthermore, each fault case of RWS and brake is considered. As a hydraulic brake system, each brake is wired diagonally. Thus, in case of brake circuit failure, the brakes cannot generate proper braking force. Considering the brake circuit failure, the faulty factors are modeled as following (15).

$$\begin{aligned}
&\textit{If Left Front and Right Rear Brake fault}, f_{1,4} = 0 \\
&\textit{If Right Front and Left Rear Brake fault}, f_{2,3} = 0
\end{aligned} \tag{15}$$

When RWS actuator motor fails in normal case, it returns to the default value, which is zero degree. However, some critical faults would make it impossible. It is difficult to overcome undesired yaw moment by a driver. Thus, fault tolerant control should be developed for driver's safety.

For coordination of RWS and ESC, the cost functions are defined as follows. The constraints of actuator and tire friction limitation are considered using KKT conditions.

1) Case 1, RWS+ESC integrated control:

$$\begin{aligned}
&\min_{F^*_{x,LF}, F^*_{x,RF}, F^*_{y_LR}} J_1 = W_f (F_{x,des} - [(f_1 + f_3 P_L) F^*_{x,LF} + (f_2 + \\
&f_4 P_R) F^*_{x,RF}])^2 + W_m [(\frac{|M_{des}|}{t} + \frac{F_{x_des}}{2}) - (f_2 + f_4 P_L) F^*_{x,RF}]^2
\end{aligned} \tag{16}$$

$$\begin{aligned}
&\textit{subject to}: \\
&f_{equal,1} : \frac{t}{2}(f_2 + f_4 P_R) F^*_{x_RF} - \frac{t}{2}(f_1 + f_3 P_L) F^*_{x_LF} \\
&\quad - l_r Q_1 F^*_{y_LR} - M_{des} = 0 \\
&g_{inequal,1} : F^*_{x,LF} \le 0,\ g_{inequal,2} : F^*_{x_RF} \le 0 \\
&g_{inequal,3} : (F^*_{x_LF})^2 + F^2_{y_LF} - \mu^2 F^2_{z_LF} \le 0 \\
&g_{inequal,4} : (F^*_{x,RF})^2 + F^2_{y,RF} - \mu^2 F^2_{z,RF} \le 0 \\
&g_{inequal,5} : (F^*_{y_LR})^2 - (2\deg \cdot \frac{\pi}{180} \cdot C_{\alpha r})^2 \le 0 \\
&\textit{where}, W_f, W_m \textit{ are weighting coefficients}
\end{aligned} \tag{17}$$

Equation (16), (17) show the objective function and constraints of the integrated chassis control of RWS and ESC. The objective function means that deceleration should be minimized to generate desired yaw moment. In other words, expected yaw moment is satisfied by RWS if the driver does not have braking intention. Also, brake circuit failure can be considered by adjusting faulty factors (15).

2) Case 2, RWS fault case:

$$\begin{aligned}
&\min_{F^*_{x,LF}, F^*_{x,RF}} J_2 = W_f (F_{x,des} - [(f_1 + f_3 P_L) F^*_{x,LF} + (f_2 + f_4 P_R) F^*_{x,RF}]^2 \\
&+ W_m [M_{des} - (-\frac{t}{2}(f_1 + f_3 P_L) F^*_{x,LF} + \frac{t}{2}(f_2 + f_4 P_R) F^*_{x,RF} \\
&- l_r Q_1 f(RWS\ fault))]^2 \\
&\textit{subject to}: \\
&g_{inequal,1} : F^*_{x,LF} \le 0, g_{inequal,2} : F^*_{x,RF} \le 0, \\
&g_{inequal,3} : (F^*_{x,LF})^2 + F^2_{y,LF} - \mu^2 F^2_{z,LF} \le 0 \\
&g_{inequal,4} : (F^*_{x,RF})^2 + F^2_{y,RF} - \mu^2 F^2_{z,RF} \le 0 \\
&\textit{where}, f(RWS\ fault) = C_{\alpha r} \cdot \delta_{rear,fault}
\end{aligned} \tag{18}$$

Equation (18) represents an objective function and constraints of RWS fault. In this case, RWS system is shut off immediately when RWS failure is detected. If RWS actuator fails to return to zero degree, the degree of RWS actuator angle is measured by the sensor. After that, the undesired yaw moment is estimated. Using estimated unintended yaw moment, ESC conducts fault tolerant control to ensure vehicle safety.

C. Optimal Solution

Lagrange functions are defined with KKT condition (19). J, λ, u and s indicate the objective function, lagrange multiplier and slack variable, respectively.

$$\begin{aligned}
L_1 &= J_1 + \lambda f_{equal,1} + \sum u_j (g_{inequal,j} + s_j^2),\ (j = 1,2,3,4,5) \\
L_2 &= J_2 + \sum u_j (g_{inequal,j} + s_j^2),\ (j = 1,2,3,4)
\end{aligned} \tag{19}$$

The first order necessary conditions are as follows (20). In this paper, optimized solution is pre-calculated and applied to the distribution logic that satisfies the necessary conditions.

$$\begin{aligned} &\frac{\partial L}{\partial F^*_{x,LF}} = \frac{\partial L}{\partial F^*_{x,RF}} = \frac{\partial L}{\partial F^*_{y,LR}} = 0 \\ &\frac{\partial L}{\partial \lambda} = \frac{\partial L}{\partial u_j} = \frac{\partial L}{\partial s_j} = 0 \end{aligned} \tag{20}$$

VI. SIMULATION RESULT

Proposed integrated chassis control was validated via closed-loop simulations with commercial software MATLAB/Simulink and Carsim.

A. Simulation Environment

In this study, the following three simulations were conducted to test the effectiveness of the presented control system. One is closed-loop simulation in Double Lane Change (DLC) and J-turn with driver model in Fig. 6. Front steering angle was determined by driver model. The others are simulations with faults. The results of the proposed control system were compared with the conventional ESC.

B. Closed-loop simulation without fault

- Desired longitudinal velocity: 70kph

The simulation under the situation without fault was performed. Fig. 7 and Fig. 8 show that integrated chassis control was well conducted according to driver's braking intention and driving situation. The following results are obtained through the experiments.

1. Depending on the driving situation, the appropriate control mode is determined. The control mode has two modes: The lateral stability mode and the handling performance mode. Side slip angle was decreased while driving in DLC. Meanwhile, yaw rate was increased while driving in the moderate driving situation, J-turn.
2. The variation of longitudinal velocity was significantly reduced.

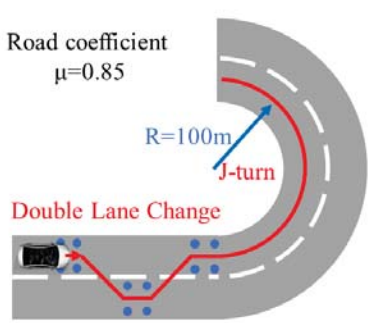

Fig. 6. Simulation scenario

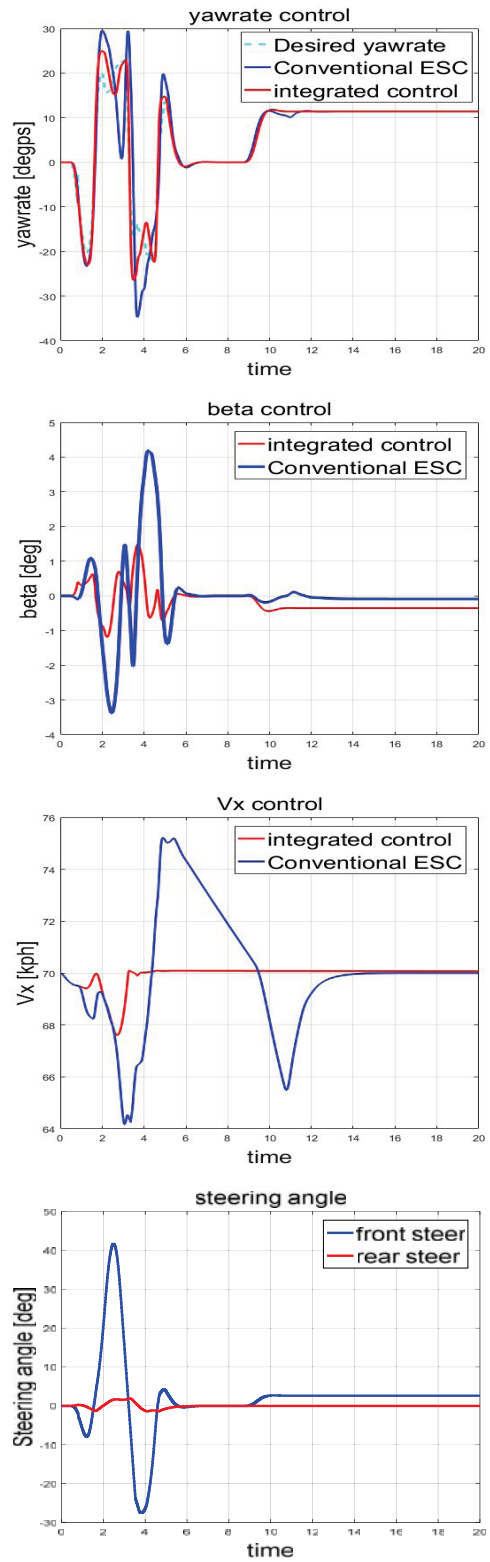

Fig. 7. Response of proposed system (a), (b), (c)

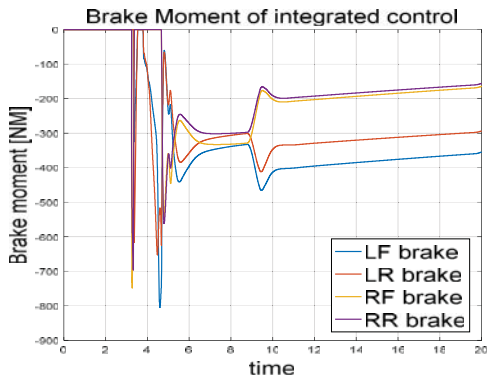

Fig. 8. Response of proposed system (d)

C. *Closed-loop Simulation with brake circuit failure*

- Fault scenario: LF, RR brake circuit failure occurs
- Longitudinal velocity: 70kph to 30kph

It is assumed that driver brakes when brake circuit has failure. Fig. 9 shows that performances of yaw rate and brake control were increased compared to conventional ESC. Also, the input of brake moment is shown in Fig. 10. In other words, the proposed integrated control was verified that the braking and handling performances were improved under brake circuit failure.

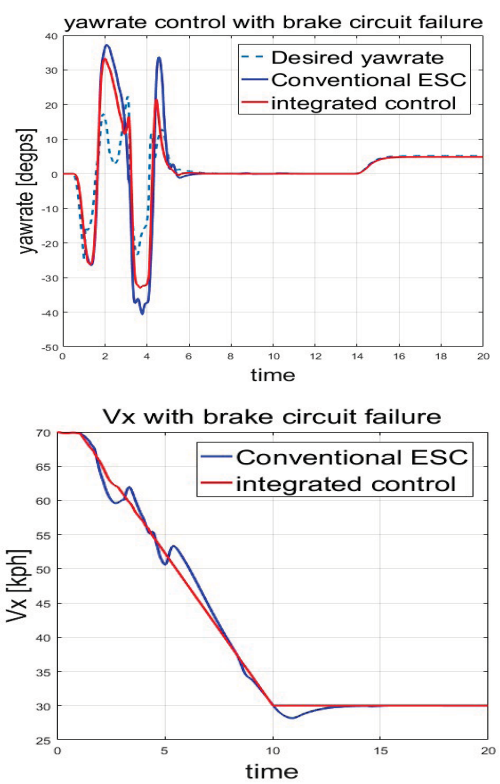

Fig. 9. Response of proposed system with brake circuit failure (a), (b)

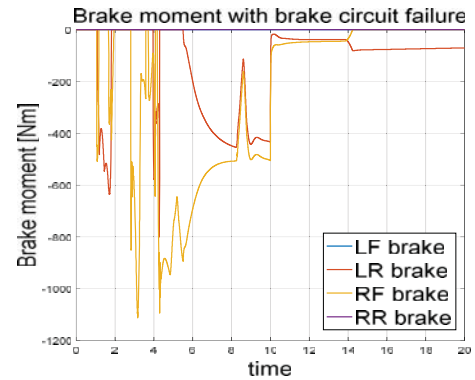

Fig. 10. Response of proposed system with brake circuit failure (c)

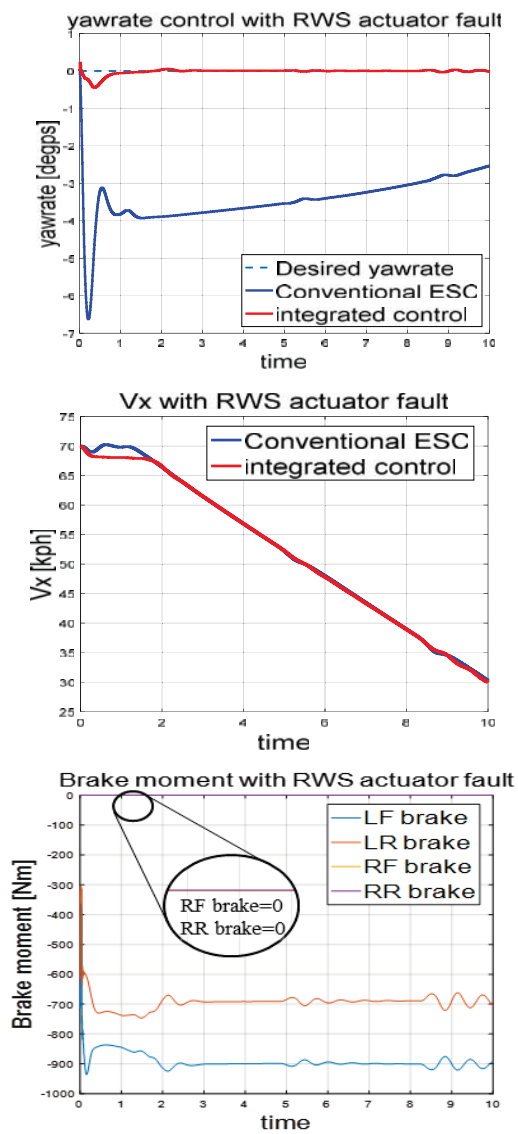

Fig. 11. Response of proposed system with RWS actuator fault (a), (b), (c)

D. Closed-loop Simulation with RWS fault

- Fault scenario: RWS actuator fault occurs at 1 degree.
- Longitudinal velocity: 70kph to 30kph
- Front steering control: 0 degree

As previously mentioned, it was assumed that RWS system is shut down immediately and the driver brakes when RWS actuator has failure. RWS actuator angle was measured by the sensor, and the unexpected yaw moment was estimated. Consequently, fault tolerant control was conducted through the optimal distributor. Fig. 11 shows that yaw rate performance was dramatically improved in RWS fault scenario compared with conventional ESC. As a result, undesired yaw moment was certainly decreased.

VII. Conclusion

In this study, integrated chassis control consisted of three parts. First was state estimator. The second was upper controller. The third was the optimal distributor. Three-axis tire forces, side slip angle and cornering stiffness were estimated in part 1. Upper controller determined vehicle driving status using vehicle stability index and target vehicle motion according to driver's intention and the driving situation in part 2. Finally, optimal coordinator distributed control input to ESC and RWS.

Presented integrated chassis control of RWS and ESC with actuator failure was investigated via MATLAB/Simulink and Carsim. The following results were obtained through the experiments.

1. Optimal coordination of RWS and ESC was performed using analytical method. As a result, handling, lateral stability and braking control performance were significantly improved.
2. Fault tolerant control was conducted using faulty factors. The simulation results showed that proposed control system had great braking stability and performance compared to conventional ESC under brake circuit failure.
3. Fault tolerant control was performed with RWS actuator fault. As a result, undesired yaw rate was effectively decreased.

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